

Waffles — Crisp Shell, Custard Middle

A shell that cracks like thin ice over a middle close to custard. Overnight yeasted batter, whipped whites folded at service — cook to the wisp of steam, then rack, never stack.

Yield: **24 waffles** (10–12) Active: **30 min** Total: **overnight** Iron: **highest** Done: **steam thins to a wisp**

WHERE IT WAS BORN

Medieval Europe — hinged irons for communion wafers and the street *oublie*, 9th–10th c. The name is Frankish *waffla*, honeycomb. The Brussels form (thin yeast batter, whipped whites) recorded by name in Ghent, 1842–43.

EATEN, AUTHENTICALLY

Standing, out of paper, under nothing but icing sugar — Brussels street and fairground food, never a plated dessert.

HOW IT'S USED NOW

A breakfast and brunch fixture worldwide since the 1964 New York World's Fair. The same batter turns savoury with cheese and pepper.

INGREDIENTS

WEIGHT (LB/OZ) · CUPS / VOLUME IN RIGHT COLUMN

THE OVERNIGHT BATTER

Plain flour	15.9 oz	3% cups
Cornflour (<i>cornstarch</i>)	5.3 oz	1¼ cups
Caster sugar	1.4 oz	3 tbsp
Fine sea salt	0.39 oz	1¼ tsp
Instant dried yeast	0.18 oz	1½ tsp
Buttermilk (<i>room temp</i>)	1 lb 9.4 oz	3 cups
Whole milk	8.5 oz	1 cup
Unsalted butter, melted	8.5 oz	1 cup + 1 tbsp
Egg yolks	6	keep whites
Vanilla extract	0.28 oz	2 tsp

AT SERVICE

Egg whites, cold	6	from above
Bicarbonate of soda	0.14 oz	¾ tsp
Butter / oil, for the plates	–	

THE RATIO — FLOUR BLEND AT 100%

Blend 100% (¼ plain · ¼ cornflour) · Dairy 160% · Butter 40% · Egg 50% · Sugar 7% · Salt 1.8% · Yeast 0.8% · Soda 0.7%. Three numbers do the crisping: the ¼ cornflour, the 40% butter, and dairy at 160% — thin enough to reach every corner of the grid.

Kit: waffle iron · scale · two large bowls · whisk · balloon whisk or mixer · wire rack · 85g ladle · lidded jug for a swell

METHOD

- Build the base — night before.** Whisk dry: 450g flour, 150g cornflour, 40g sugar, 11g salt, 5g yeast. Whisk in 720g buttermilk + 240g milk until smooth, then 240g melted butter, 6 yolks, vanilla. It should ribbon off the whisk.

Overnight the yeast makes acids and aroma no chemical leavening gives, and the flour hydrates fully so the batter flows into the grid instead of sitting on it. — McGee
- Ferment, then chill.** 1 hr at ambient until it just bubbles, then fridge **8–24 hr**. Hot galley: straight to the fridge.
- Stand & preheat.** Batter out **45 min** so the butter loosens. Iron on its highest setting a **full 10 min**.

The shell is built by conduction, so plate temperature at the pour decides the crust. A plate still climbing steams the batter instead of drying it.
- Soda in, whites folded.** Whisk 4g soda through. Whip 6 whites to **soft peaks** — not stiff — and fold in three. Stop while streaky.

The foam is a gas phase built in advance: open cells that survive into the crumb. The soda hits the buttermilk acid for a fast surge, and lifts pH, which speeds browning. — Ajandouz & Puigserver 1999
- Brush, pour, close, leave it.** Brush both plates. About **85g** per waffle into the centre, short of the edges — it spreads. Do not open to look.
- Cook to the wisp. 4–6 min.** Hard steam at first, then a thin wisp — that is the cue, not the timer and never the light.

While the plume is heavy the crust is still wet and capped at 100°C. When it thins, the crust has dropped below ~9–10% water into the glassy, brittle state — only then can it brown. — Roudaut et al. 2002
- Rack it. Never stack it.** Straight onto a wire rack, single layer, uncovered, in a **100°C** oven.

Done: the steam drops from a hard jet to a thin wisp; the lid lifts without resistance and the waffle releases itself; it sounds hollow tapped on the rack; deep chestnut, not blond. **Single layer, uncovered** — trapped steam turns brittle to leather in about a minute.

Efficiency & service — the night-before batter makes service a five-minute job: soda in, whites folded, pour. Two irons and one long rack; the iron is the bottleneck, never the batter. Freeze the surplus flat and toast from frozen — waffles come back better than they hold.

ONE BATTER, TWO DISHES — THE SAVOURY WAFFLE

After the fold, **divide the bowl**. Into one half work **150g grated Gruyère**, cracked black pepper and a scrape of nutmeg; serve under a **fried egg**, or split with ham and béchamel. *Drop the iron one setting — cheese against the plates catches fast.*

ELEVATION

TIER 1 — NO EXTRA TIME

- Icing sugar only** — the Brussels way
- Brown the butter** — same 240g, hazelnut
- Malt powder** — 5% of the flour

TIER 2 — WORTH IT

- Beer** — for 20% of the dairy; crisper
- Sourdough starter** — 150g for the yeast
- Buckwheat** — 15% of the flour

TIER 3 — SERVICE

- Macerated strawberries** + crème fraîche
- Gruyère + fried egg** — the crew tier
- Split** with ham and béchamel

TROUBLESHOOT

Soft in minutes	Stacked or covered → wire rack, single layer, uncovered
Pale and blond	Iron not fully preheated, or the soda was missed → a full 10 min
Sticks & tears	Dry plates, or opened early → brush every time; leave it shut
Dense crumb	Whites stiff, or folded smooth → soft peaks, stop while streaky
Crisp then limp	Poured too thick → thinner pour, hotter iron

CHARTER PREP & STORAGE

Best route	Batter the night before; soda and whites at service only
Batter	24 hr fridge max — past that it goes beery
Hold	Wire rack, single layer, 95–100°C oven, up to 30 min
Frozen	Cool flat, bag; 200°C 5–6 min from frozen, or the toaster
Never	Stack hot, cover, or hold in a warming box